

		$= 0.127 \text{ J (0.13 J)}$	1
4	(a)	By the principle of conservation of energy, $KE_{Xi} + KE_{Yi} + GPE_{Xi} + GPE_{Yi} = KE_{Xf} + KE_{Yf} + GPE_{Xf} + GPE_{Yf} + WD \text{ against friction}$ $0 + 0 + 0 + 0 = KE_{Xf} + KE_{Yf} + (4.0)(9.81)(3.0 \sin 30^\circ) + (5.0)(9.81)(-3.0) + (10.0)(3.0)$ $KE_{Xf} + KE_{Yf} = 58 \text{ J (or 58.3 J)}$ * Correct calculation of change in GPE of X and change in GPE of Y [M1] * Correct substitution in COE equation [M1]	2 1
	(b)	Since X and Y are connected by an inextensible cord, both bodies attain the same final speed. $\frac{1}{2}(4.0 + 5.0)v^2 = 58.3$ $v_X = v_Y = 3.6 \text{ ms}^{-1}$	1

5	(a)	(i)	The centripetal force is provided by the gravitational force acting on stars	1
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